

Supervised Fine-Tuning for Colloquial Tamil Register

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https://github.com/edithandrews/tamil_diglossia

Abstract

Tamil exhibits diglossia: Literary Tamil (LT) dominates written and formal text, while Spoken/Colloquial Tamil (ST) is used in everyday speech. Because most Tamil NLP resources and multilingual MT training data come from written sources, many models default to formal or literary Tamil even in conversational contexts. This paper investigates whether colloquial register can be transferred from a large commercial teacher model (Sarvam Mayura) into a smaller student model (mBART-50) using supervised fine-tuning on teacher-generated synthetic data. Four fine-tuning regimes—IA3 adapters, LoRA, QLoRA, and full fine-tuning—are compared against a zero-shot baseline on 3,000 synthetic training examples, evaluated on 100 held-out IruMozhi test examples using a colloquial Tamil classifier. LoRA and full fine-tuning both exceed the teacher’s own colloquial classification rate while IA3 performs at the level of the zero-shot baseline, despite using more parameters. LoRA achieves strong register transfer using approximately 1.5% of trainable parameters. A secondary decoder-only probe (Qwen2.5) under the same LoRA regime reaches comparable classifier scores, suggesting the result generalizes across architectures. Register is evaluated separately from adequacy using chrF against the teacher output and qualitative inspection, showing that colloquial-looking output can still be semantically incorrect.

1 Introduction

Tamil is a classical Dravidian language with approximately 80 million native speakers. Its diglossic structure is long-documented in the

descriptive literature: Literary Tamil (LT) is the written, formal register, while Spoken Tamil (ST) is the everyday conversational register. The two varieties differ substantially in morphology, verb forms, and vocabulary. Because LT dominates NLP training corpora, multilingual models tend to produce formal Tamil even in conversational contexts.

This paper studies English-to-colloquial-Tamil translation as a register-transfer problem: the goal is not only to generate Tamil, but to generate Tamil that is recognizably ST.

The central research question is: *Which fine-tuning regime most effectively transfers modern-colloquial Tamil register from a large teacher model into a smaller student model?* The experiment is framed as supervised fine-tuning (SFT) on teacher-generated synthetic targets, with Sarvam Mayura (Sarvam AI, 2026) as the teacher and mBART-large-50 as the student. This project follows a training-focus track: the primary contribution is a systematic comparison of fine-tuning regimes for register transfer, rather than dataset construction or novel evaluation methodology.

2 Related Work

Tamil diglossia and evaluation resources. Tamil diglossia has been extensively documented in the descriptive linguistics literature (Schiffman, 1998, 1999). From an NLP perspective, the primary resource for LT/ST distinction is IruMozhi (Prasanna and Arora, 2024), which provides 499 Tamil sentence pairs with parallel literary and colloquial annotations alongside a released xlm-roberta-base-irumozhi classifier. Because IruMozhi is designed for diglossia research rather than clean English-to-Tamil MT, it is most useful here as a register benchmark rather than a

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translation evaluation resource. Tamil MT more broadly remains a low-resource problem: most available parallel corpora (e.g., IndicCorp, Tatoeba) reflect written Tamil and do not cover the colloquial register.

Style and register transfer. Text style transfer—rewriting text in a target style while preserving meaning—is a well-studied problem in NLP (Shen et al., 2017; John et al., 2019). Formality transfer, the closest analogue to this work, has been studied for English (Rao and Tetreault, 2018) and a handful of other languages, but not for Tamil diglossia specifically. A key distinction is that this paper frames the task as translation rather than rewriting: the input is English and the target is colloquial Tamil, so there is no source-language style to preserve. The register signal comes entirely from the training targets.

Knowledge distillation and teacher-generated targets. The training setup is related to sequence-level knowledge distillation (Kim and Rush, 2016), in which a student is trained on translations generated by a larger teacher rather than on human references. This approach has been shown to reduce the complexity of the target distribution and improve student training efficiency. Unlike soft-label distillation (Hinton et al., 2015), the student here is trained on hard teacher outputs only, with no access to teacher logits or intermediate representations. The teacher (Sarvam Mayura) is a black-box commercial API, making soft distillation infeasible.

Parameter-efficient fine-tuning. Full fine-tuning of large pretrained models is expensive and prone to catastrophic forgetting. Parameter-efficient fine-tuning (PEFT) methods address this by updating only a small subset of parameters. Adapter layers insert small trainable modules between frozen transformer layers. LoRA (Hu et al., 2022) instead reparametrizes weight updates as low-rank matrix products, keeping the base model frozen and adding roughly 1–2% trainable parameters. QLoRA (Dettmers et al., 2023) extends LoRA to quantized base models, enabling training on commodity hardware. IA3 (Liu et al., 2022) is an extreme case: it learns only element-wise scaling vectors

applied to keys, values, and feedforward activations, yielding $\sim 0.03\%$ trainable parameters. Theoretically, IA3’s expressivity is more limited than LoRA’s, since it cannot model additive shifts in the representation space—only multiplicative rescaling. This distinction is relevant here, since register transfer may require more than rescaling existing Tamil-register representations.

Multilingual MT and mBART. mBART-50 (Tang et al., 2020) is a multilingual denoising autoencoder pretrained on 50 languages, including Tamil. It is used as the student model because it provides a strong multilingual prior and generates Tamil natively, avoiding the need for transliteration or script conversion at inference time. The fine-tuning regimes compared are full fine-tuning, LoRA, QLoRA, and IA3.

3 Data

IruMozhi. The IruMozhi dataset (Prasanna and Arora, 2024) serves as the primary evaluation resource and one source of training examples. Of the 499 available examples, 399 are used for training and 100 are held out for evaluation. Each training example pairs the English source sentence with the corresponding Sarvam Mayura modern-colloquial translation as the target, ensuring the student learns to match the teacher’s colloquial style.

Teacher pipeline. Sarvam Mayura (Sarvam AI, 2026) (mayura:v1) is used as the teacher translation model. Sarvam exposes three register modes: formal, classic-colloquial, and modern-colloquial. A sanity check confirms that the modern-colloquial mode achieves 96.8% colloquial classification by the IruMozhi classifier, close to the 99.2% rate at which the IruMozhi human colloquial references themselves are classified as colloquial. The human-reference rate is not a true human ceiling, since the classifier was trained on these labels and so is expected to score near 100% on its own training distribution, it is reported only as an upper anchor for comparison. This mode is used exclusively as the training target. Sarvam outputs are postprocessed into a normalized romanized form.

Tatoeba augmentation. To supplement the 399 IruMozhi training examples, 3,166 English sentences from the Tatoeba English–Tamil corpus (Tatoeba Project, 2026) are translated using Sarvam Mayura modern-colloquial mode, then filtered to retain examples classified as colloquial by the IruMozhi classifier. The final mixed3000 training set contains 399 IruMozhi + 2,601 Tatoeba teacher examples (3,000 total). The IruMozhi classifier is used both to filter training data and to evaluate student outputs, so the reported % colloquial figures should be interpreted as measuring alignment with the training distribution rather than register quality. A different classifier might produce different numbers. The training filter is justified by the goal of training on data that is reliably colloquial in the first place.

Output postprocessing. The Sarvam modern-colloquial mode returns code-mixed Tamil that combines Tamil script with English-script tokens (e.g., “1908-ல launch ஆச்சு”). A postprocessing script converts the Tamil-script portions to the Aksharamukha romanization scheme and then applies a normalization pass to bring the output into a more standard romanized form (“1908-la launch aachu”). This step is required so that chrF and BLEU scores computed against the IruMozhi human references are not artificially lowered solely due to differences in script and orthography mismatch.

Data quality limitations. IruMozhi presents systematic challenges for MT evaluation. The English column contains awkward or overly literal phrasings that are not well-suited to natural translation. The human colloquial reference appears to be a literary-to-colloquial conversion rather than a direct translation from English, introducing differences in form that make reference comparison difficult. Additionally, human references use informal Latin romanization (e.g., *avan*), while Sarvam and mBART outputs use Aksharamukha/ISO conventions (e.g., *avan*), artificially lowering chrF and BLEU scores even when the outputs are semantically identical.

4 Methods

4.1 Student Model

The student model is facebook/mbart-large-50-many-to-many-mmt (Tang et al., 2020), a 0.6B parameter multilingual sequence-to-sequence transformer pretrained on 50 languages. The tokenizer is configured with `src_lang=en_XX` and `tgt_lang=ta_IN`, and forced BOS (Beginning of Sequence) decoding uses the Tamil language token. Input sequences are truncated to 128 tokens, and output generation uses beam search with beam width 4.

4.2 Training Regimes

Four fine-tuning regimes are compared against a zero-shot baseline, selected to span a wide range of parameter efficiency and to test a specific hypothesis about expressivity: if register transfer requires introducing new representational directions (not just rescaling existing ones), then IA3 should fail while LoRA succeeds. Full fine-tuning serves as the no-constraint upper bound; QLoRA tests whether quantization degrades register transfer under the same parameter budget as LoRA.

Zero-shot baseline. No fine-tuning is performed. The pretrained mBART-50 model generates Tamil directly from English prompts. This baseline establishes the default model register without any colloquial training signal and is reported alongside the four fine-tuning regimes for comparison.

Full fine-tuning. All model parameters are updated during training. This represents the upper bound for parameter investment: all 0.6B parameters of mBART-50 are trainable.

LoRA. Low-Rank Adaptation (Hu et al., 2022) is applied with rank $r = 32$, scaling factor $\alpha = 64$ (so the effective learning rate of the adapter is $\alpha/r = 2$), dropout $p = 0.1$, and target modules `q_proj`, `k_proj`, `v_proj`, and `out_proj`. This yields approximately 1.5% trainable parameters relative to full model size. The LoRA adapter is injected into both encoder and decoder attention layers; all other weights remain frozen. The rank $r = 32$ was chosen as a middle point between expressivity and parameter count; to assess sensitivity, ablations are additionally trained at $r =$

16 ($\sim 0.8\%$ trainable parameters) and $r = 8$ ($\sim 0.4\%$ trainable parameters).

QLoRA. LoRA (same configuration as above) is applied on top of a base model loaded in 4-bit NF4 quantization using bitsandbytes (Dettmers et al., 2023). This enables training on more memory-constrained hardware while maintaining roughly the same trainable parameter count as standard LoRA.

IA3 (adapter). Infused Adapter by Inhibiting and Amplifying Inner Activations (Liu et al., 2022) is applied using the PEFT library’s IA3Config with target modules `k_proj`, `v_proj`, and `fc2`. IA3 introduces element-wise scaling vectors rather than additive weight matrices, yielding approximately 0.03% trainable parameters—roughly $30\times$ fewer than LoRA.

4.3 Training Setup

All fine-tuned models share the following hyperparameters: 3 epochs of training on mixed3000, batch size 4, gradient accumulation over 2 steps (effective batch size 8), fp16 mixed-precision training, and gradient checkpointing. The AdamW optimizer is used with default parameters. Training is performed on a single NVIDIA A100 40GB GPU via Google Colab Pro. All runs use a single fixed random seed with no validation set or early stopping; hyperparameters were not tuned on the test set.

4.4 Evaluation Pipeline

All four fine-tuning regimes and the zero-shot baseline are evaluated on 100 held-out IruMozhi test examples. The test set is small: with binary outcomes at $n = 100$, a single misclassified example shifts the % colloquial figure by one percentage point. For each generated output, the pipeline does the following:

1. Applies the IruMozhi classifier (aryaman/xlm-roberta-base-irumozhi, released alongside the IruMozhi dataset to obtain a binary colloquial/literary label and confidence score.
2. Aggregates % colloquial and mean classifier score across 100 examples.
3. Computes chrF and BLEU against the human IruMozhi colloquial reference as

separate adequacy proxies, using sacreBLEU (Popovic, 2015; Papineni et al., 2002) with default settings (BLEU 13a tokenization; CHRF $n = 6$). Both scores are reported on the standard 0–100 scale.

4. Computes an ad-hoc modernity index, introduced in this paper as a style diagnostic independent of the human reference: for a model output o with chrF score m against the Sarvam modern-colloquial anchor and c against the classic-colloquial anchor, the modernity index is

$$\text{Modernity}(o) = \frac{m}{m + c}$$

averaged across the 100 examples. Values near 1 indicate similarity to the modern-colloquial anchor; values near 0 indicate similarity to the classic-colloquial anchor.

The teacher model (Sarvam Mayura modern-colloquial) is included as a reference condition.

This separation is important: the classifier answers whether an output looks like Spoken Tamil, while chrF/BLEU and manual audit examples ask whether it is an adequate translation of the source.

5 Results

The results fall into three groups. LoRA reaches 97% colloquial classification and full fine-tuning 96%, both slightly exceeding the 93% teacher rate and representing an improvement of roughly 84 percentage points over the 13% zero-shot baseline. QLoRA forms a middle group at 91%, somewhat below LoRA and the teacher but still well within the colloquial range. IA3 and the zero-shot baseline sit in a third group at 15% and 13% respectively. The very small IA3 configuration used here, with only $\sim 0.03\%$ trainable parameters, does not transfer a meaningful colloquial register signal in this setup and behaves essentially like the untrained model.

The modernity index confirms the classifier result: full, LoRA, and QLoRA outputs score around 0.64–0.65, well above IA3 and baseline at 0.39–0.40. It does not, however, separate full fine-tuning from LoRA at the aggregate level.

Table 1: Register classification results on 100 IruMozhi test examples.

Condition	% Colloq.	Modernity
Teacher (Sarvam)	93%	—
Full fine-tuning	96%	0.642
LoRA	97%	0.652
QLoRA	91%	0.642
IA3 (adapter)	15%	0.404
Zero-shot baseline	13%	0.385

Table 2: chrF scores against two references: human-annotated IruMozhi colloquial references and the Sarvam teacher outputs.

Condition	chrF	
	vs. Human Ref	vs. Teacher
Full fine-tuning	26.09	66.97
LoRA ($r = 32$)	23.41	59.05
LoRA ($r = 16$)	23.53	52.33
LoRA ($r = 8$)	23.49	49.33
QLoRA	23.20	56.71
IA3 (adapter)	27.04	22.93
Zero-shot baseline	26.77	20.65

Against the human reference, chrF does not track the register ranking: IA3 and the baseline score slightly higher than LoRA and full fine-tuning, because the formal outputs are more similar to the literary-to-colloquial references. The absolute values are low across the board, and BLEU values cluster near zero for all conditions and provide essentially no signal. When student outputs are instead scored against the Sarvam teacher outputs directly, the ranking sharpens considerably and aligns with the classifier result: full fine-tuning is closest to the teacher, followed by LoRA and QLoRA, with the rank ablations ($r = 16$: 52.3, $r = 8$: 49.3) stepping down monotonically, while IA3 and the zero-shot baseline fall far below. This dissociation suggests that fine-tuned models learn the teacher’s phrasing successfully, but diverge from the human references because those references were produced by converting literary Tamil rather than translating from English directly. Reference-overlap scores against human annotations are thus consistent with the register findings, but given the reference quality issues, they cannot be taken as independent evidence of transla-

tion adequacy. Comparing student outputs directly against the teacher is a more meaningful diagnostic of how well each fine-tuning regime approximates the target register.

To assess sensitivity to training data volume, LoRA is also trained on three smaller subsets: irumozhi399 (399 examples), mixed1000, and mixed2000. Table 3 shows that chrF_T improves monotonically with data volume (45.0 → 51.9 → 55.1 → 59.1), while the classifier score remains high ($\geq 96\%$) across all settings. This confirms that the classifier saturates early—register transfer is achievable even with 399 examples—but surface fidelity to the teacher continues to improve with more data, consistent with the register/adequacy distinction discussed in Section 6.

Table 3: LoRA data scaling results. % Colloq and chrF_T on 100 IruMozhi test examples.

Training set	% Colloq	chrF _T
irumozhi399	96%	45.0
mixed1000	96%	51.9
mixed2000	97%	55.1
mixed3000	97%	59.1

As a secondary probe, two decoder-only models were fine-tuned under the same LoRA regime on both the irumozhi399 and mixed3000 training sets: Qwen2.5-0.5B-Instruct and Qwen2.5-1.5B-Instruct ((Qwen Team, 2025)). The zero-shot baselines fail in different ways: the 1.5B model produces Tamil-sounding but incoherent output (18% colloquial), while the 0.5B model produces code-mixed gibberish that the classifier rates higher (50%) simply because it contains more English tokens. After LoRA fine-tuning, both models reach high colloquial classifier scores across both data settings (Table 5). Qualitative inspection shows that outputs trained on irumozhi399 alone are often semantically inadequate despite high register scores, whereas mixed3000-trained outputs are noticeably more adequate. These results suggest that register transfer via LoRA generalizes across encoder-decoder and decoder-only architectures, but that data volume remains the limiting factor for adequacy even when the register signal is strong.

Table 4: Three illustrative qualitative examples. Outputs are in romanized Tamil.

Condition	“I try to explain as far as I know.”	“Launched in 1908.”	“The tail is also white.”
Teacher	enakkuth therinja varaikkum naan explain panren.	1908-la launch aachu.	tail-um white-aa irukkum.
Full	enakkuth therinja varaikkum naan explain panna try panren.	1908-la launch pannappattathu.	vaayilyum white-aa irukku.
LoRA	enakkuth teriyum alavirku explain panna try pannaren.	1908-la launch pannaanga.	tail-um white-aayum irukku.
IA3	enakkuth terintavarai naan vilakka muyalkiren.	1908-l thodangappattathu.	ivarrin vaalkalum venmaiyaanavai.
Baseline	enakkuth terintavarai naan vilakka muyarsikkiren.	1908 il niruvappattathu.	ivarrin muttaiyum venmaiyaanathu.

Table 5: Decoder-only probe results. chrF_T : chrF against the Sarvam teacher output.

Model	Training	% Colloq	chrF_T
Teacher	—	0.93	—
Qwen-1.5B	Zero-shot	0.18	17.1
	LoRA (399)	0.96	43.0
	LoRA (3000)	0.97	53.6
Qwen-0.5B	Zero-shot	0.50	19.3
	LoRA (399)	0.99	36.9
	LoRA (3000)	0.96	39.3

6 Analysis and Discussion

6.1 Qualitative Register Transfer

Table 4 presents three illustrative examples comparing generated outputs across conditions, alongside the teacher reference.

The examples illustrate several features of colloquial register transfer. In the first example, full fine-tuning and LoRA both preserve the meaning and use modern-colloquial mixed vocabulary around *explain pan-*, while IA3 and the baseline preserve the meaning in more literary Tamil. The second example shows a style distinction: LoRA uses the more colloquial active construction *launch pannaanga*, while full fine-tuning uses the more formal passive *pannappattathu*. The third example shows why register and adequacy must be kept separate: the full model produces a colloquial-looking output (*vaayilyum white-aa irukku*) but uses *vaayil* (locative of *vaay*, “mouth”) where the correct word is *vaal* (tail). LoRA avoids the lexical error by code-switching to English, which the classifier still reads as colloquial.

6.2 Fine-Tuning Regime Analysis

The three-group result raises a natural question: why does IA3 fail while LoRA succeeds? The most likely explanation is expressivity. IA3 applies element-wise rescaling to keys, values, and feedforward activations, which can suppress or amplify existing Tamil register features but cannot introduce additive shifts in representation space. If the pre-trained mBART-50 model has not internalized colloquial Tamil patterns during pretraining—plausible given the LT bias of available Tamil text—then rescaling existing representations may be insufficient to produce them. LoRA, by contrast, adds low-rank weight updates that can represent new directions in the activation space, making it better suited to transferring a register that the base model underrepresents. The QLoRA gap (91% vs. 97% for LoRA) is smaller and likely attributable to quantization noise during backpropagation: 4-bit NF4 quantization introduces rounding error that may slightly degrade the quality of gradient updates, particularly for a task that depends on fine-grained lexical choices.

The rank ablation results in Table 2 show a clear monotonic trend in chrF_T : $r = 32$ (59.1) $> r = 16$ (52.3) $> r = 8$ (49.3), while the classifier scores remain high across all three ranks (97%, 90%, 85%). This dissociation suggests that low-rank adapters can capture enough of the register signal to satisfy the classifier even at $r = 8$, but that higher rank is needed to match the teacher’s specific phrasing. Register transfer is a coarser objective than surface-form imitation: a model can produce colloquial-looking Tamil with limited parameters, but reproducing the teacher’s exact lexical choices requires more capacity.

6.3 Register, Adequacy, and Data Quality

A finding of this work is that register transfer and translation adequacy can come apart. The classifier metric reflects surface register features: colloquial morphology, spoken-style pronouns, and mixed-script lexical choices. It does not measure whether the translation is semantically faithful to the English source. The third column of Table 4 (“The tail is also white”) makes this concrete: the full model generates a translation which has colloquial morphology but uses the wrong word. This is not merely a metric limitation, it is a substantive result about what register-only training optimizes for. A model fine-tuned on a register signal can produce output that conforms to the register, but is nonetheless semantically incorrect, and high classifier confidence is necessary but not sufficient for translation quality.

The data quality issues documented in Section 3—awkward English sources, literary-to-colloquial references, and romanization mismatch—further constrain interpretation of chrF and BLEU scores; both are therefore reported as secondary diagnostics only. The held-out test set of 100 examples is sufficient for stable classifier estimates but limits the precision of surface overlap comparisons. This is treated as a limitation of the evaluation setup rather than a flaw in the experimental design, since the central claim of this work is specifically about register transfer.

7 Conclusion

This paper investigated colloquial Tamil register transfer from the Sarvam Mayura teacher model into mBART-50 using supervised fine-tuning on 3,000 synthetic training examples, comparing four fine-tuning regimes against a zero-shot baseline on 100 held-out IruMozhi test examples. The main finding is that LoRA reaches 97% colloquial classification and full fine-tuning 96%—both slightly above the 93% teacher rate—while IA3 reaches only 15%, essentially identical to the 13% zero-shot baseline. LoRA achieves this result using only ~1.5% of trainable parameters, indicating that LoRA-scale parameter efficiency is sufficient for register transfer in this setup. A decoder-

only probe under the same LoRA regime reaches comparable classifier scores (97% and 96% respectively on mixed3000), with the 1.5B model closing further on the teacher in chrF_T (53.6 vs. 39.3). This suggests that register transfer via LoRA generalizes across encoder-decoder and decoder-only architectures.

The central caveat is that classifier-recognized register transfer does not guarantee translation adequacy. Qualitative inspection exposes cases where an output looks colloquial but is semantically incorrect, which is a result of optimizing for a register signal rather than for meaning. Future work should strengthen this separation along several axes. First, a dedicated human adequacy annotation pass would give a direct measure of translation quality independent of register. Second, a small lexicon of diagnostic literary vs. colloquial forms (locatives, copulas, present-tense morphology, code-switched light verbs) would provide an interpretable register signal that is methodologically independent of the IruMozhi classifier and does not inherit its training-data biases. Third, the evaluation would benefit from datasets with cleaner English sources and matched romanization conventions, so that chrF and BLEU can serve as informative adequacy proxies rather than noisy diagnostics.

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